

Micro Lending Architecture Justification

PES2UG24CS036 | Micro-Lending and Peer Credit Risk Assessor

Architecture Selection

We chose **Layered Architecture for the Micro-Lending and Peer Credit Risk Assessor System**. The system manages sensitive financial profiles, credit-risk assessment, repayment scheduling, repayment recording, default warnings, authorization, and transaction integrity. Presentation, application-service, and security/data layers give these responsibilities clear boundaries while preserving one reliable transaction model.

Architectural Style Analysis

Style	Fit for this scenario	Main tradeoff
Layered	Strong fit: separates financial workflows while keeping one ACID data boundary.	Some cross-layer call overhead.
Microservices	Could scale risk and repayment independently, but distribution complicates atomic updates.	Network, deployment, and consistency complexity.
Client-Server	Centralizes rules and data, but provides weaker internal separation by itself.	Server bottleneck and single point of failure.

Two Scenario Specific Reasons

1 Reliable financial transactions. Repayment recording must preserve ACID behavior. The Repayment Service uses a repository transaction interface so the payment record, outstanding balance, and related loan state can commit or roll back together inside one data-layer boundary.

2 Maintainable workflow separation. Credit-risk rules, repayment logic, authorization, and persistence change for different reasons. Stable interfaces let each component evolve and be tested independently while the Loan Management Service coordinates the complete lending workflow.

Security Advantage

Protected financial data. The Authorization Service authenticates the user and returns a role-and-permission decision before the Loan Management Service performs protected profile or loan operations. The client never accesses the repository directly. This centralizes policy enforcement, narrows the sensitive-data boundary, and supports least-privilege access.

Performance Benefit

Low-latency internal processing. Risk assessment, loan coordination, authorization, and repayment logic can communicate through local service interfaces instead of multiple network hops. This keeps profile validation and repayment updates responsive, while stable risk-policy data can be cached without bypassing the repository as the source of truth.

Component and Interface Summary

Six components are used: Borrower and Lender Portal, Loan Management Service, Credit Risk Engine, Repayment Service, Authorization Service, and Financial and Loan Repository. Their six assembly interfaces cover loan management, risk, repayment, authorization, profile data, and atomic repayment transactions.